

**TP LSQL pour DAI – L2 – EMIT FIANARANTSOA (2025-2026)**

**A: QCM**

1. De quoi SQL est-il l'acronyme ?

- ☐ Strong Query Language
- ☐ Strong Question Language
- ☐ Structured Question Language
- ☒ Structured Query Language

2. Quel mot-clé SQL est utilisé pour rechercher des données dans une base de données ?

- ☐ FIND
- ☐ GET
- ☐ SEARCH
- ☒ SELECT

3. Quel mot-clé SQL est utilisé pour modifier des données dans une base de données ?

- ☐ MODIFY
- ☐ REPLACE
- ☒ UPDATE
- ☐ CHANGE

4. Quel mot-clé SQL est utilisé pour ajouter de nouvelles données dans une base de données ?

- ☐ ADD
- ☐ APPEND
- ☒ INSERT
- ☐ RECORD

5. Quel mot-clé SQL est utilisé pour supprimer des données dans une base de données ?

- ☐ DELETE
- ☐ SUPPRESS
- ☐ CANCEL
- ☒ REMOVE

6. Quel(s) mot(s)-clé(s) SQL est utilisé pour trier les résultats d'une requête ?

- ☐ ORDER
- ☒ ORDER BY
- ☐ SORT
- ☐ SORT BY

1) JEU OLYMPIQUE

```
CREATE DATABASE medail;
PRAGMA foreign_keys=OFF; SET FOREIGN_KEY_CHECKS=0;
START TRANSACTION
CREATE TABLE medals(
  Id INT AUTO_INCREMENT PRIMARY KEY,
  City VARCHAR(10) ,
  Year INT(4) ,
  Sport VARCHAR(10) ,
  Discipline VARCHAR(255) ,
  Event VARCHAR(40) ,
  Athlete VARCHAR(100) ,
  Gender VARCHAR(50) ,
  Country_code VARCHAR(50) ,
  Country VARCHAR(50) ,
  Event_gender VARCHAR(50) ,
  Medal VARCHAR(20)
);
```

a.

## SQL

```
SELECT *
FROM Medals
WHERE Year=2008 AND Country="France";
```

| Id     | City    | Year         | Sport    | Discipline | Event                  | Athlete             | Gender | Country_cod |
|--------|---------|--------------|----------|------------|------------------------|---------------------|--------|-------------|
| e      | Country | Event_gender | Medal    |            |                        |                     |        |             |
| 13453  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | GILOT, Fabien       | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13458  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | LEVEAUX, Amaury     | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13478  | Beijing | 2008         | Aquatics | Swimming   | 50m freestyle          | BERNARD, Alain      | Men    | FRA         |
| France | M       |              | Bronze   |            |                        |                     |        |             |
| 13507  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | STEIMETZ, Boris     | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13515  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | BOUSQUET, Frederick | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13573  | Beijing | 2008         | Aquatics | Swimming   | 100m freestyle         | BERNARD, Alain      | Men    | FRA         |
| France | M       |              | Gold     |            |                        |                     |        |             |
| 13577  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | MALLET, Gregory     | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13579  | Beijing | 2008         | Aquatics | Swimming   | 50m freestyle          | LEVEAUX, Amaury     | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13582  | Beijing | 2008         | Aquatics | Swimming   | 4x100m freestyle relay | BERNARD, Alain      | Men    | FRA         |
| France | M       |              | Silver   |            |                        |                     |        |             |
| 13615  | Beijing | 2008         | Aquatics | Swimming   | 200m breaststroke      | DUBOSCQ, Hugues     | Men    | FRA         |
| France | M       |              | Bronze   |            |                        |                     |        |             |

|        |         |      |            |                 |                                        |                            |       |     |
|--------|---------|------|------------|-----------------|----------------------------------------|----------------------------|-------|-----|
| 14876  | Beijing | 2008 | Rowing     | Rowing          | quadruple sculls without coxswain (4x) | COEFFIC, Jonathan          | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14886  | Beijing | 2008 | Rowing     | Rowing          | four without coxswain (4-)             | RONDEAU, Benjamin          | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14890  | Beijing | 2008 | Rowing     | Rowing          | quadruple sculls without coxswain (4x) | PELTIER, Pierre-Jean       | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14910  | Beijing | 2008 | Rowing     | Rowing          | four without coxswain (4-)             | DESPRES, Julien            | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14915  | Beijing | 2008 | Rowing     | Rowing          | four without coxswain (4-)             | CHARDIN, Germain           | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14968  | Beijing | 2008 | Rowing     | Rowing          | quadruple sculls without coxswain (4x) | BERREST, Cedric            | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 14986  | Beijing | 2008 | Rowing     | Rowing          | quadruple sculls without coxswain (4x) | BAHAIN, Julien             | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 15024  | Beijing | 2008 | Sailing    | Sailing         | 470 - Two Person Dinghy                | BAUSSET, Olivier           | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 15028  | Beijing | 2008 | Sailing    | Sailing         | 470 - Two Person Dinghy                | CHARBONNIER, Nicolas       | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 15030  | Beijing | 2008 | Sailing    | Sailing         | RS:X - Windsurfer                      | BONTEMPS, Julien           | Men   | FRA |
| France | M       |      | Silver     |                 |                                        |                            |       |     |
| 15045  | Beijing | 2008 | Sailing    | Sailing         | Finn - Heavyweight Dinghy              | FLORENT, Guillaume         | Men   | FRA |
| France | X       |      | Bronze     |                 |                                        |                            |       |     |
| 15072  | Beijing | 2008 | Shooting   | Shooting        | skeet (125 targets)                    | TERRAS, Anthony            | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 15205  | Beijing | 2008 | Taekwondo  | Taekwondo       | 57 - 67 kg                             | EPANGUE, Gwladys Patience  | Women | FRA |
| France | W       |      | Bronze     |                 |                                        |                            |       |     |
| 15361  | Beijing | 2008 | Weightlift | Weightlifting   | 69kg                                   | DABAYA-TIENTCHEU, Vencelas | Men   | FRA |
| France | M       |      | Silver     |                 |                                        |                            |       |     |
| 15429  | Beijing | 2008 | Wrestling  | Wrestling Gre-R | 66 - 74kg                              | GUENOT, Christophe         | Men   | FRA |
| France | M       |      | Bronze     |                 |                                        |                            |       |     |
| 15432  | Beijing | 2008 | Wrestling  | Wrestling Gre-R | 60 - 66kg                              | GUENOT, Steeve             | Men   | FRA |
| France | M       |      | Gold       |                 |                                        |                            |       |     |

76 rows in set (0.044 sec)

b.

## SQL

```
SELECT City, Year, Athlete
FROM Medals
WHERE Medal="Gold" AND Event="110m hurdles";
```

```
SELECT City , Year,Athlete FROM Medals WHERE Medal="Gold" AND Event="110m hurdles";
```

| City       | Year | Athlete         |
|------------|------|-----------------|
| Montreal   | 1976 | DRUT, Guy       |
| Moscow     | 1980 | MUNKELT, Thomas |
| Los Angele | 1984 | KINGDOM, Roger  |
| Seoul      | 1988 | KINGDOM, Roger  |
| Barcelona  | 1992 | MCKOY, Mark     |
| Atlanta    | 1996 | JOHNSON, Allen  |
| Sydney     | 2000 | GARCIA, Anier   |
| Athens     | 2004 | LIU, Xiang      |
| Beijing    | 2008 | ROBLES, Dayron  |

9 rows in set (0.008 sec)

C.

```
MariaDB [medail]> SELECT Athlete,Event FROM Medals WHERE Medal="Gold" AND Country="France" AND Year=2000 ORDER BY Athlete ASC;
```

| Athlete                  | Event                         |
|--------------------------|-------------------------------|
| ANDRIEUX, Michel         | coxless pair (2-)             |
| ASLOUM, Brahim           | - 48kg (light-flyweight)      |
| BALLANGER, Felicia       | sprint                        |
| BALLANGER, Felicia       | 500m time trial               |
| BETTE, Jean-Christophe   | lightweight coxless four (4-) |
| DORFMANN, Xavier         | lightweight coxless four (4-) |
| DOUILLET, David          | + 100kg (heavyweight)         |
| DUMOULIN, Franck         | 10m air pistol (60 shots)     |
| ESTANGUET, Tony          | C-1 (canoe single)            |
| FERRARI, Jean-Noel       | foil team                     |
| GANE, Laurent            | Olympic Sprint                |
| GUYART, Brice            | foil team                     |
| HOCDE, Yves              | lightweight coxless four (4-) |
| LHOTELLIER, Patrice      | foil team                     |
| MARTINEZ, Miguel         | cross-country                 |
| PLUMENAIL, Lionel        | foil team                     |
| PORCHIER, Laurent        | lightweight coxless four (4-) |
| ROLLAND, Jean Christophe | coxless pair (2-)             |
| ROUSSEAU, Florian        | Olympic Sprint                |
| ROUSSEAU, Florian        | Keirin                        |
| TOURNANT, Arnaud         | Olympic Sprint                |
| VANDENHENDE, Severine    | 57 - 63kg (half-middleweight) |

22 rows in set (0.010 sec)

d.

```
MariaDB [medail]> SELECT DISTINCT Country FROM Medals WHERE Medal="Gold" AND Year="1984" ORDER BY country DESC;
```

| Country        |
|----------------|
| Yugoslavia     |
| West Germany   |
| United States  |
| United Kingdom |
| Sweden         |
| Spain          |
| Romania        |
| Portugal       |
| Pakistan       |
| New Zealand    |
| Netherlands    |
| Morocco        |
| Mexico         |
| Korea, South   |
| Kenya          |
| Japan          |
| Italy          |
| France         |
| Finland        |
| China          |
| Canada         |
| Brazil         |
| Belgium        |
| Austria        |
| Australia      |

25 rows in set (0.012 sec)

e.

```
MariaDB [medail]> SELECT Athlete, City ,Year ,Medal,Country FROM Medals WHERE Country LIKE "%nia";
```

| Athlete            | City     | Year | Medal  | Country |
|--------------------|----------|------|--------|---------|
| MEGELEA, Gheorghe  | Montreal | 1976 | Bronze | Romania |
| DAFINOIU, Costica  | Montreal | 1976 | Bronze | Romania |
| SIMON, Mircea      | Montreal | 1976 | Silver | Romania |
| ZILBERMAN, Victor  | Montreal | 1976 | Bronze | Romania |
| CUTOV, Simion      | Montreal | 1976 | Silver | Romania |
| NASTAC, Alexandru  | Montreal | 1976 | Bronze | Romania |
| DIBA, Vasile       | Montreal | 1976 | Bronze | Romania |
| DIBA, Vasile       | Montreal | 1976 | Gold   | Romania |
| SERHEI, Larion     | Montreal | 1976 | Bronze | Romania |
| SIMIONOV, Gheorghe | Montreal | 1976 | Silver | Romania |
| MALIHIN, Policarp  | Montreal | 1976 | Bronze | Romania |
| DANIELOV, Gheorghe | Montreal | 1976 | Silver | Romania |
| NILCA, Alexandru   | Montreal | 1976 | Bronze | Romania |
| POP, Ioan          | Montreal | 1976 | Bronze | Romania |
| MARIN, Corneliu    | Montreal | 1976 | Bronze | Romania |
| IRIMICIUC, Daniel  | Montreal | 1976 | Bronze | Romania |
| MUSTATA, Marin     | Montreal | 1976 | Bronze | Romania |

|                            |         |      |        |           |
|----------------------------|---------|------|--------|-----------|
| IZBASA, Sandra             | Beijing | 2008 | Gold   | Romania   |
| GRIGORE, Andreea           | Beijing | 2008 | Bronze | Romania   |
| ACATRINEI, Andreea         | Beijing | 2008 | Bronze | Romania   |
| DUMITRU, Alina Alexandra   | Beijing | 2008 | Gold   | Romania   |
| POLAVDER, Lucija           | Beijing | 2008 | Bronze | Slovenia  |
| KRUNGOLCAS, Edvinas        | Beijing | 2008 | Silver | Lithuania |
| ZADNEPROVSKIS, Andrejus    | Beijing | 2008 | Bronze | Lithuania |
| BARABAS, Eniko             | Beijing | 2008 | Bronze | Romania   |
| SUSANU, Viorica            | Beijing | 2008 | Bronze | Romania   |
| IGNAT, Doina               | Beijing | 2008 | Bronze | Romania   |
| SUSANU, Viorica            | Beijing | 2008 | Gold   | Romania   |
| JAANSON, Juri              | Beijing | 2008 | Silver | Estonia   |
| PAPUC, Ioana Cristina      | Beijing | 2008 | Bronze | Romania   |
| GEORGESCU, Elena           | Beijing | 2008 | Bronze | Romania   |
| ANDRUNACHE, Georgeta       | Beijing | 2008 | Gold   | Romania   |
| ANDRUNACHE, Georgeta       | Beijing | 2008 | Bronze | Romania   |
| ENDREKSON, Tonu            | Beijing | 2008 | Silver | Estonia   |
| SERBAN, Rodica Maria       | Beijing | 2008 | Bronze | Romania   |
| MUSAT, Simona Dumitrita    | Beijing | 2008 | Bronze | Romania   |
| BURCICA, Constanta         | Beijing | 2008 | Bronze | Romania   |
| ZBOGAR, Vasilij            | Beijing | 2008 | Silver | Slovenia  |
| VOLUNGEVICIUTE, Gintare    | Beijing | 2008 | Silver | Lithuania |
| DEBEVEC, Rajmond           | Beijing | 2008 | Bronze | Slovenia  |
| DAVTYAN, Gevorg            | Beijing | 2008 | Bronze | Armenia   |
| MARTIROSYAN, Tigran Varban | Beijing | 2008 | Bronze | Armenia   |
| MARTIROSYAN, Tigran Gevorg | Beijing | 2008 | Bronze | Armenia   |
| MIZGAITIS, Mindaugas       | Beijing | 2008 | Bronze | Lithuania |
| AMOYAN, Roman              | Beijing | 2008 | Bronze | Armenia   |
| PATRIKEEV, Yuri            | Beijing | 2008 | Bronze | Armenia   |

577 rows in set (0.010 sec)

2)

a) La liste *sans doublons* des noms de tous les athlètes français ayant obtenu une médaille d'or aux jeux olympiques de 1984.

```
MariaDB [medail]> SELECT DISTINCT Athlete FROM Medals WHERE Country = 'France' AND Medal = 'Gold' AND Year = 1984;
```

| Athlete                    |
|----------------------------|
| QUINON, Pierre             |
| LAMOUR, Jean-Francois      |
| BOISSE, Philippe           |
| BIBARD, Michel             |
| RUST, Albert               |
| GARANDE, Patrice           |
| ZANON, Jean Louis          |
| BRISSON, Francois          |
| CUBAYNES, Patrick          |
| BENSOUSSAN, Michel         |
| THOUVENEL, Jean Christophe |
| LACOMBE, Guy               |
| LEMOULT, Jean-Claude       |
| AYACHE, William            |
| BIJOTAT, Dominique         |
| SENAC, Didier              |
| JEANNOL, Philippe          |
| TOURE, Jose                |
| XUEREB, Daniel             |
| ROHR, Jean Philippe        |
| HEBERLE, Philippe          |

21 rows in set (0.010 sec)

b) Les noms de toutes les athlètes féminines ayant obtenu une médaille d'or au marathon et l'année d'obtention de cette médaille, avec un classement chronologique.

```
MariaDB [medail]> SELECT Athlete, Year
-> FROM Medals
-> WHERE Gender = 'Women' AND Medal = 'Gold' AND Event = 'marathon'
-> ORDER BY Year ASC;
```

| Athlete              | Year |
|----------------------|------|
| BENOIT, Joan         | 1984 |
| MOTA, Rosa           | 1988 |
| YEGOROVA, Valentina  | 1992 |
| ROBA, Fatuma         | 1996 |
| TAKAHASHI, Naoko     | 2000 |
| NOGUCHI, Mizuki      | 2004 |
| TOMESCU, Constantina | 2008 |

7 rows in set (0.009 sec)

c) La liste de tous les athlètes qui ont le même prénom que vous, avec la nature de leur médaille et leur discipline (la requête doit se terminer par *WHERE Athlete LIKE* "%Pierre%" si votre prénom est Pierre).

```
MariaDB [medail]> SELECT Athlete, Medal, Discipline FROM Medals WHERE Athlete LIKE '%Pierre%';
```

| Athlete                | Medal  | Discipline   |
|------------------------|--------|--------------|
| QUINON, Pierre         | Gold   | Athletics    |
| GUICHOT, Pierre        | Silver | Fencing      |
| DELOOF, Pierre Marie   | Silver | Rowing       |
| DURAND JR., Pierre     | Bronze | Jumping      |
| DURAND JR., Pierre     | Gold   | Jumping      |
| FAIVRE-PIERRET, Didier | Bronze | Cycling Road |
| GUICHOT, Pierre        | Bronze | Fencing      |
| THORSSON, Pierre       | Silver | Handball     |
| THORSSON, Pierre       | Silver | Handball     |
| AMAT, Jean Pierre      | Bronze | Shooting     |
| AMAT, Jean Pierre      | Gold   | Shooting     |
| PIERRE, Gabriel        | Silver | Baseball     |
| WOME NLEND, Pierre     | Gold   | Football     |
| THORSSON, Pierre       | Silver | Handball     |
| PELTIER, Pierre-Jean   | Bronze | Rowing       |

```
15 rows in set (0.008 sec)
```

d) Les épreuves dans lesquelles le champion de natation américain [Michael Phelps](#) a obtenu des médailles en 2004.

```
MariaDB [medail]> SELECT Event FROM Medals WHERE Athlete = 'Michael Phelps' AND Year = 2004;
Empty set (0.006 sec)
```

e. Le nombre total de médailles d'or et d'argent obtenues par l'Australie en natation.

```
MariaDB [medail]> SELECT COUNT(*) FROM Medals WHERE Country = 'Australia' AND Discipline = 'Swimming' AND Medal IN ('Gold', 'Silver');
```

| COUNT(*) |
|----------|
| 133      |

```
1 row in set (0.009 sec)
```

3)

```
MariaDB [medail]> SELECT DISTINCT Country FROM Medals WHERE Medal="Gold" AND Year="1984" ORDER BY country DESC;
```

| Country        |
|----------------|
| Yugoslavia     |
| West Germany   |
| United States  |
| United Kingdom |
| Sweden         |
| Spain          |
| Romania        |
| Portugal       |
| Pakistan       |
| New Zealand    |
| Netherlands    |
| Morocco        |
| Mexico         |
| Korea, South   |
| Kenya          |
| Japan          |
| Italy          |
| France         |
| Finland        |
| China          |
| Canada         |
| Brazil         |
| Belgium        |
| Austria        |
| Australia      |

```
25 rows in set (0.012 sec)
```

## Prix Nobel

1.

```
CREATE TABLE IF NOT EXISTS nobel (
  Id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  Year INT,
  Category VARCHAR(100),
  Prize VARCHAR(255),
  Motivation TEXT,
  PrizeShare VARCHAR(50),
  LaureateID INT,
  LaureateType VARCHAR(50),
  FullName VARCHAR(255),
  BirthDate VARCHAR(50),
  BirthCity VARCHAR(255),
  BirthCountry VARCHAR(100),
  Sex VARCHAR(20),
  OrganizationName VARCHAR(255),
  OrganizationCity VARCHAR(255),
  OrganizationCountry VARCHAR(100),
  DeathDate VARCHAR(50),
  DeathCity VARCHAR(255),
  DeathCountry VARCHAR(100)
) ENGINE=InnoDB;
```

2.

a)



```
MariaDB [prix_nobel]> SELECT DISTINCT Category FROM nobel;
```

| Category   |
|------------|
| Chemistry  |
| Literature |
| Medicine   |
| Peace      |
| Physics    |
| Economics  |

```
6 rows in set (0.002 sec)
```

b)

```
MariaDB [prix_nobel]> SELECT DISTINCT FullName, Year FROM nobel WHERE BirthCountry = 'France' OR OrganizationCountry = 'France' ORDER BY FullName AS
```

| FullName                       | Year |
|--------------------------------|------|
| Alain Aspect                   | 2022 |
| Albert Fert                    | 2007 |
| Alexis Carrel                  | 1912 |
| Alfred Kastler                 | 1966 |
| Alfred Werner                  | 1913 |
| Anatole France                 | 1921 |
| André Frédéric Cournand        | 1956 |
| André Lwoff                    | 1965 |
| André Paul Guillaume Gide      | 1947 |
| Anne L'Huillier                | 2023 |
| Annie Ernaux                   | 2022 |
| Antoine Henri Becquerel        | 1903 |
| Aristide Briand                | 1926 |
| Charles Edouard Guillaume      | 1920 |
| Charles Jules Henri Nicolle    | 1928 |
| Charles Louis Alphonse Laveran | 1907 |

|                                                                                       |      |
|---------------------------------------------------------------------------------------|------|
| Léon Jouhaux                                                                          | 1951 |
| Léon Victor Auguste Bourgeois                                                         | 1920 |
| Marie Curie, née Skłodowska                                                           | 1911 |
| Martin Karplus                                                                        | 2013 |
| Maurice Allais                                                                        | 1988 |
| Michel Devoret                                                                        | 2025 |
| Moungi Bawendi                                                                        | 2023 |
| Patrick Modiano                                                                       | 2014 |
| Paul Henri Benjamin Balluet d'Estournelles de Constant, Baron de Constant de Rebecque | 1909 |
| Paul Sabatier                                                                         | 1912 |
| Philippe Aghion                                                                       | 2025 |
| Pierre Agostini                                                                       | 2023 |
| Pierre Curie                                                                          | 1903 |
| Pierre-Gilles de Gennes                                                               | 1991 |
| Prince Louis-Victor Pierre Raymond de Broglie                                         | 1929 |
| René Cassin                                                                           | 1968 |
| Roger Guillemin                                                                       | 1977 |
| Roger Martin du Gard                                                                  | 1937 |
| Romain Rolland                                                                        | 1915 |
| Serge Haroche                                                                         | 2012 |
| Seán MacBride                                                                         | 1974 |
| Sully Prudhomme                                                                       | 1901 |
| Victor Grignard                                                                       | 1912 |
| Yves Chauvin                                                                          | 2005 |

```
72 rows in set (0.002 sec)
```

c)

```
MariaDB [prix_nobel]> SELECT Year FROM nobel WHERE FullName = 'Comité international de la Croix Rouge';
Empty set (0.002 sec)
```

d)

```
MariaDB [prix_nobel]> SELECT COUNT(*) FROM nobel WHERE Sex = 'Female';
+-----+
| COUNT(*) |
+-----+
|        69 |
+-----+
row in set (0.001 sec)
```

e)

```
MariaDB [prix_nobel]> SELECT FullName, Category FROM nobel WHERE Sex = 'Female' AND BirthCountry = 'France' AND Year >= 2005;
+-----+-----+
| FullName | Category |
+-----+-----+
| Françoise Barré-Sinoussi | Medicine |
| Emmanuelle Charpentier | Chemistry |
| Annie Ernaux | Literature |
| Esther Duflot | Economics |
| Anne L'Huillier | Physics |
+-----+-----+
5 rows in set (0.001 sec)
```

f)

```
MariaDB [prix_nobel]> SELECT FullName, (2026 - CAST(LEFT(BirthDate, 4) AS UNSIGNED)) AS Age FROM nobel WHERE DeathDate IS NULL AND LaureateType = 'Individual' ORDER BY Age DESC;
+-----+-----+
| FullName | Age |
+-----+-----+
| Paul D. Boyer | 108 |
| Jens C. Skou | 108 |
| Nicolaas Bloembergen | 106 |
| Edmond H. Fischer | 106 |
| Jack Steinberger | 105 |
| Kenneth J. Arrow | 105 |
| Chen Ning Yang | 104 |
| Stanley Cohen | 104 |
| Leon M. Lederman | 104 |
| Hans G. Dehmelt | 104 |
| Rudolph A. Marcus | 103 |
| Henry A. Kissinger | 103 |
| Arvid Carlsson | 103 |
| Philip Warren Anderson | 103 |
+-----+-----+
```

g)

```
MariaDB [prix_nobel]> SELECT FullName FROM nobel WHERE FullName LIKE '%Curie%';
+-----+
| FullName |
+-----+
| Pierre Curie |
| Marie Curie, née Skłodowska |
| Marie Curie, née Skłodowska |
| Irène Joliot-Curie |
+-----+
4 rows in set (0.001 sec)
```

3. a-

```
MariaDB [prix_nobel]> INSERT INTO nobel (Year, Category, LaureateType, FullName, BirthCountry, Sex) VALUES (2026, 'Hacking', 'Individual', 'Sandrah Alegne', 'Madagascar', 'Female');
Query OK, 1 row affected (0.003 sec)

MariaDB [prix_nobel]> SELECT * FROM nobel
-> WHERE FullName = 'Sandrah Alegne';
```

| Id   | Year | Category | Prize            | Motivation       | PrizeShare          | LaureateID | LaureateType | FullName       | BirthDate | BirthCity | BirthCountry | Sex |
|------|------|----------|------------------|------------------|---------------------|------------|--------------|----------------|-----------|-----------|--------------|-----|
|      |      |          | OrganizationName | OrganizationCity | OrganizationCountry | DeathDate  | DeathCity    | DeathCountry   |           |           |              |     |
| 1087 | 2026 | Hacking  | NULL             | NULL             | NULL                | NULL       | Individual   | Sandrah Alegne | NULL      | NULL      | Madagascar   | Fem |
| ale  | NULL |          | NULL             | NULL             | NULL                | NULL       | NULL         | NULL           |           |           |              |     |

b-

```
MariaDB [prix_nobel]> UPDATE nobel SET Category = 'NSI' WHERE FullName = 'Sandrah Alegne';
Query OK, 1 row affected (0.020 sec)
Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [prix_nobel]> SELECT FullName, Category, Year
-> FROM nobel
-> WHERE FullName = 'Sandrah Alegne';
```

| FullName       | Category | Year |
|----------------|----------|------|
| Sandrah Alegne | NSI      | 2026 |

1 row in set (0.001 sec)

c-

```
MariaDB [prix_nobel]> DELETE FROM nobel WHERE FullName = 'Sandrah Alegne';
Query OK, 1 row affected (0.005 sec)

MariaDB [prix_nobel]> SELECT FROM nobel WHERE FullName = 'Sandrah Alegne';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'FROM nobel WHERE FullName = 'Sandrah Alegne'' at line 1
MariaDB [prix_nobel]> SELECT * FROM nobel WHERE FullName = 'Sandrah Alegne';
Empty set (0.002 sec)
```

## LES JOUEURS DE BASKET ÉVOLUANT EN NBA

1 . CREATE DATABASE nba;

CREATE TABLE IF NOT EXISTS joueurs\_nba (

    Name VARCHAR(255),

```

Team VARCHAR(255),

Number INT,

Position VARCHAR(50),

Age INT,

Height INT,

Weight INT

) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

2. Cette requête permet de calculer l'âge moyen des joueurs de l'équipe des Los Angeles Lakers

```

MariaDB [nba]> SELECT AVG(Age)FROM joueurs_nba WHERE Team = "Los Angeles Lakers";
+-----+
| AVG(Age) |
+-----+
| 26.1429 |
+-----+
1 row in set (0.001 sec)

```

3.a)

```

MariaDB [nba]> SELECT MAX(Height) FROM joueurs_nba WHERE Team = 'Chicago Bulls';
+-----+
| MAX(Height) |
+-----+
| 211 |
+-----+
1 row in set (0.001 sec)

```

b)

```

MariaDB [nba]> SELECT MIN(Weight) FROM joueurs_nba WHERE Position = 'SG';
+-----+
| MIN(Weight) |
+-----+
| 75 |
+-----+
1 row in set (0.001 sec)

```

c)

```
MariaDB [nba]> SELECT * FROM joueurs_nba WHERE Name = 'Victor Wembanyama';
```

| Name              | Team              | Number | Position | Age | Height | Weight |
|-------------------|-------------------|--------|----------|-----|--------|--------|
| Victor Wembanyama | San Antonio Spurs | 1      | F        | 21  | 221    | 107    |

```
1 row in set (0.001 sec)
```

Pour savoir s'il est le plus jeune des joueurs de NBA de plus de 2m10

```
MariaDB [nba]> SELECT Name, Age, Height FROM joueurs_nba WHERE Height > 210 ORDER BY Age ASC LIMIT 1;
```

| Name           | Age | Height |
|----------------|-----|--------|
| Khaman Maluach | 19  | 218    |

```
1 row in set (0.001 sec)
```

#### 4. Modification de la base de données

```
MariaDB [nba]> INSERT INTO joueurs_nba (Name, Team, Number, Position, Age, Height, Weight) VALUES ('Sandrah Alegne', 'Boston Celtics', 99, 'PF', 20, 205, 100);
```

Query OK, 1 row affected (0.013 sec)

```
MariaDB [nba]> SELECT * FROM joueurs_nba WHERE Name = 'Sandrah Alegne';
```

| Name           | Team           | Number | Position | Age | Height | Weight |
|----------------|----------------|--------|----------|-----|--------|--------|
| Sandrah Alegne | Boston Celtics | 99     | PF       | 20  | 205    | 100    |

```
1 row in set (0.001 sec)
```